Measurement: Sensors 31 (2024) 100945

Contents lists available at ScienceDirect

Measurement: Sensors

journal homepage: www.sciencedirect.com/journal/measurement-sensors

PCA based feature extraction and MPSO based feature selection for gene expression microarray medical data classification

Abdul Razzaque [a], Dr Abhishek Badholia [b,*]

[a] CSE Department, MATS University, Raipur, C.G, India
[b] CSE Department, MATS University, Raipur, C.G, India

ARTICLE INFO

Keywords:
Support vector machine (SVM)
Naïve bayesian
KNN
PCA
PSO
Medical data

ABSTRACT

In this paper, a novel Multi Class based Feature Extraction (MC-FE) method has been proposed for medical data classification. Genomic datasets, or gene expression-based microarray medical datasets, are categorised for cancer diagnosis. The first stage involves applying a feature extraction technique. The Principal Component Analysis (PCA) is used to extract the features for medical data classification to detect leukemia, colon tumors, and prostate cancer. The MPSO (modified particle swarm optimization) technique is used at the second stage to pick features from high-dimensional microarray medical datasets like prostate cancer, leukemia, and colon tumors. Finally, SVM, KNN, and Naive Bayes classifiers are used to classify medical data. Then, the classified data are stored in the cloud IOT sensors. When compared to the existing methods, the proposed method gives better optimized results for SVM, KNN, and Naive Bayes of 0.88, 0.86, and 0.73, respectively. The results of cancer data extraction and feature selection are also contrasted and assessed using certain performance measure factors.

1. Introduction

Through microarray technology, experimental molecular biology has made significant contributions to the medical decision support system over the previous two decades. Microarrays aid in the storage of a cell's gene expression level. Cancer is linked to the genome, and oncogenes, which act as transducers to send signals to the nucleus, are linked to the disease. Gene expression profiles, according to molecular studies are unique to each disease. High dimensionality, short or relatively small sample size, considerable variation in non-random noise atmosphere, sample backgrounds, high redundancy, and nonlinearity are typical properties of gene expression profiles. The design of accurate and efficient classifiers, and the evaluation of the classifiers' performance are the major tasks in microarray classification. The microarray technology is prominently employed in determining the gene expression profiles of diseases and illness treatment which mandates the selection of a miniscule number of genes from galore, and its expression values are utilized to generate classification profiles. This technique gene selection method is tricky as it requires mapping of limited samples to abundantly available genes [1–3]. The main contribution of this study,

- The Principal Component Analysis (PCA) is used to extract features for medical data classification to detect prostate cancer, leukemia, and colon tumors.
- The MPSO (modified particle swarm optimization) technique is employed at the second stage to pick features from high-dimensional microarray medical datasets like leukemia, colon tumors, and prostate cancer.
- Finally, SVM, KNN, and Naive Bayes classifiers are used to classify medical data. Then, the classified data are stored in the cloud IOT sensors.

The remaining parts of this study are already set up as follows: section 2 summarizes the related works, section 3 explains the suggested approach, section 4 recounts the findings of simulations and measurements, and section 5 contains the conclusion.

2. Literature review

Occurrence of several new varieties of cancer has affected in modification of gene bank and has resulted in expressions of gene bank running in megabytes. This clearly presents a challenge to the machine learning, bioinformatics, data mining and soft computing research

* Corresponding author.
E-mail addresses: ab.razzaque@rediffmail.com (A. Razzaque), abadholia23@gmail.com (D.A. Badholia).

https://doi.org/10.1016/j.measen.2023.100945
Received 5 January 2023; Received in revised form 10 May 2023; Accepted 12 November 2023
Available online 18 November 2023
2665-9174/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

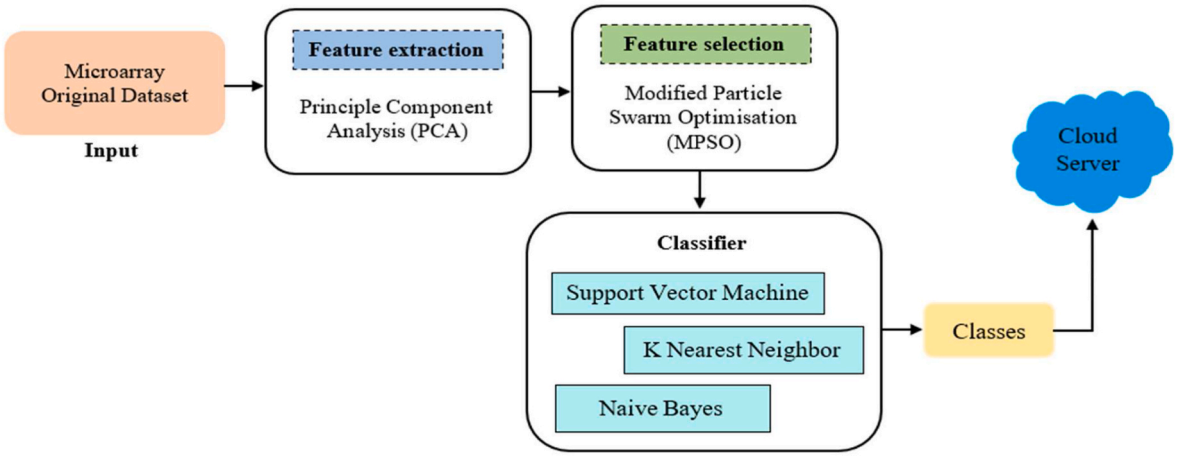


Fig. 1. The overall block diagram of the proposed model.

communities in terms of clustering, prediction, classification and drug development [4–6].

Microarrays aid in the storage of gene expression levels in a cell, helping to transform the gene's data into a final gene product. To produce gene expression data with considerable problems as mentioned, several high throughput techniques are used. Ontogeny serves as a transducer, connecting with the nucleus, and demonstrates a direct link between the sickness and cancer. Cancer is linked to the genome.

Each disease displays a different gene expression profile, according to molecular studies thus rendering the primary focus of microarray analysis as cancer categorization. Classifier creation needs a reduced and efficient data set which is achieved through extracting feature and selecting feature under microarray classification. There are two types of cancer classification: binary and multiclass. Lung cancer, Brain tumour, CNS, NCI60, Leukemia and SRBCT are classified into binary and multiclass classification category.

In 2020 Aasam Abdul Karim et al. [7] observed the inefficiency of MPSO (Modified Particle Swarm Optimization) to handle complex optimization problem as MPSO is very critical regarding the fitness information of global best position and not caring the diversity information of swarm. Their aim was to improve search performance of algorithm that handles the optimization problem with different characteristics. They used OGC (Optimal Guide Creation) module into MPSO EG by referring the particle with directional information which is valuable. Global exemplar and Local exemplar were the by product of OGC module with Global exemplar being employed to guide the swarm converging to the promising solution region.

Zahid Ansari et al. [8] analysed the medical data using different classification techniques. They covered analysis of three hard computing algorithms viz. KNN, Naive Bayes & Decision tree and three soft computing algorithms viz. SVM, Fuzzy k-means clustering and ANN. The dataset used were Breast Cancer Wisconsin, contraceptive Method Choice and Haberman data. In conferred by them that soft computing algorithm has good performance over traditional hard computing algorithms.

Vinay A. et al. [9,10] developed a novel method that used ORB (Oriented Fast and Rotated Brief) to extract features before applying PCA for dimension reduction. They employed SURF PCA (Speeded Up Robust Feature), SIFT PCA (Scale invariant Feature transform), ORB and ORB-PCA methodologies, among other analysis techniques. They asserted that their method was more quickly and precisely.

Up to this point, it is clear that various linear and nonlinear classifiers have been proposed in the literature. There are several applications that use classifiers including the Naive Bayes classifier, K-nearest Neighbor (KNN), Support vector machines (SVM), CART, random forest, and decision tree classifier. Within the group of classifiers, researchers have chosen SVM as the most frequently and comprehensively used technique. SVMs of the first generation could only categorise cancers into binary classes, but more recently developed SVMs, such as DAGSVM and the Crammer and Singer (CS) approach, can now classify tumors into several classes, which is more useful and accurate for cancer diagnosis [11–13].

Table 1a
Bifurcation of total samples used for classification.

Dataset	Prostate Cancer	Colon Tumor	Leukemia
Class	2	2	2
Total Samples	136	62	72
Training Samples	77	40	50
Test Samples	59	22	22
Original Features	12600	2000	7129

The biomarkers which carry important information relevant to cancer are mined out to significantly reduce the dataset's dimension. Gene expression is profiled using a variety of feature mining methods. Hybridization of features is carried out to create new ones using feature extraction methods [14]. Computational Intelligence is employed to analyse microarray data. Various classifiers are such as including SVM, KNN, and Naïve Bayesian [15–18] have been put to work for Cancer classifications. While respecting the dataset's originality, the feature mining process optimizes the feature subset from the input space.

Feature selection procedures are filtering, wrapping, and embedding. Wrapper strategies seek the best feature set and classifiers simultaneously, whereas filter techniques focus solely on feature subset derivation and neglect classifiers [19–22]. Feature selection is done during the training phase for embedded techniques. There is a large array of feature mining techniques employed for a huge dataset with IOT. In the above-mentioned existing techniques, only one dataset is used for classification. But in our proposed method, there are three types of medical datasets used for classification.

3. Proposed approach

Feature extraction, feature selection, and the building of robust and

Table 1b
Summary of study inputs, techniques and measures.

Dataset	Prostate Cancer, Leukemia, Colon Tumor
Feature extraction	Principle Component Analysis (PCA)
Feature Selection	Modified Particle Swarm Optimization (MPSO)
Classifier used	SVM, KNN, and Naive Bayes
Performance evaluation	Total accuracy, precision, recall, F-score, AUC and ROC

Table 2
Feature extraction after implementing PCA.

Dataset	#Classes	#Samples	#Original Features	Features reduced	% Reduced
Prostate Cancer	2	270	12600	70	0.55 %
Colon Tumour	2	62	2000	60	3 %
Leukemia	2	72	7129	143	2 %

Table 3
Training samples & test samples used for classification.

Dataset	#Training samples	#Testing samples
Prostate Cancer	203	67
Colon Tumour	43	19
Leukemia	50	22

efficient classifiers are the three key phases in microarray dataset categorization. The medical data classification is proposed in this paper [23–25]. Gene Expression micro array medical data are classified after making certain analysis. After analyzing the pros and cons of the available methods in the literature it is suggested to use PCA (Principal Component Analysis) to extract feature and MPSO (Modified Particle Swarm optimization) to select feature respectively. Finally, SVM, KNN, and Naive Bayes classifiers are used to classify medical data. Then, the classified data are deposited in the cloud IOT sensors. Fig. 1 illustrated the overall block diagram of the proposed model.

3.1. Description of datasets used in the study

The microarray medical dataset is online available and it is used in the work. The URL is (http://datam.i2r.a-star.edy.sg/datasets/krbd). The data for prostate cancer, colon tumors, and leukemia are taken from the Kent Ridge Biomedical Datasets repository and used in the suggested research [26,27].

(A.) **Prostate Cancer:** There are around 12600 gene expression levels of 136 samples in prostate cancer. In this dataset, only two numbers of classes are there. Out of these 136 samples 77 are used for training and 59 are used for testing. From 77 samples 52 are tumour and 25 are non-tumour. Similarly, from 59 samples 50 and 9 are tumour and non-tumour, respectively [28].

(B.) **Colon Tumor:** From colon tumour patient 62 samples including 40 '-ve' i.e., tumour biopsies and 22 '+ve' i.e., normal healthy biopsies from the same sufferer are considered. From 6500 features only 2000 are selected

(C.) **Leukemia:** Acute Myeloid Leukemia (AML) and Acute Lymphoblastic Leukemia (AML) are the two different kinds of leukemia. This dataset contains total 72 samples including 47 of 'ALL' and 25 of AML. Total gene expressions are 7129.

Principal Component Analysis (PCA) is the feature extraction method used in the research. On the dataset mentioned above, viz., Modified Particle Swarm Optimization (MPSO) is used to select the feature. Colon cancer, leukemia, and prostate cancer. Three alternative machine learning classifiers such as Support Vector Machine (SVM), K Nearest Neighbor (KNN), and Naive Bayes were used to validate the assertion [29,30]. SVM, KNN, and Naive Bayes are the most widely used machine learning algorithms for classification. SVM is a machine learning method for supervised learning that can be used to solve classification- or regression-related problems. The most common application, nevertheless, is in classification issues. The Naive Bayes algorithm, on the other hand, is a classification method based on the Bayes rule and a number of conditional independence assumptions. While both KNN and SVM algorithms successfully classify images, they accomplish so with varying degrees of accuracy. Compared to the KNN, the SVM offers much higher classification accuracy and speed. To evaluate the performance Precision, recall, F-score, total accuracy, ROC and AUC are obtained [31–34]. These are summarised in below Table 1b.

3.1.1. PCA for feature extraction for medical data classification

Significant features from original microarray datasets are extracted using Principal Component Analysis (PCA), Kernel PCA [26,27] and Linear Discriminate Analysis (LDA). PCA is used to extract features from Colon Tumour and Prostate Cancer microarray datasets in this study [35–38]. PCA is a linear transform tool working on data such that the first coordinates reflect highest variance, the second coordinates displaying second highest variance and so on. Algorithm 1 shows the pseudocode for PCA.

Table 4(a)
Results of prostate cancer.

Dataset	#Classes	Classifier	Recall	Precision	AUC	F-Score	Calculated Accuracy (Overall)
Prostate cancer	2	SVM	0.88	0.91	0.904	0.88	0.880
		Naive Bayesian	0.73	0.81	0.872	0.68	0.865
		KNN	0.87	0.90	0.892	0.87	0.731

Table 4(b)
Results of colon tumour.

Dataset	#Classes	Classifier	Recall	Precision	AUC	F-Score	Cal. Accuracy (Overall)
Colon Tumor	2	SVM	0.84	0.89	0.870	0.85	0.884
		Naive Bayesian	0.79	0.80	0.789	0.78	0.801
		KNN	0.84	0.84	0.842	0.84	0.750

Table 4(c)
Results of leukemia.

Dataset	#Classes	Classifier	Recall	Precision	AUC	F-Score	Calculated Accuracy (Overall)
Leukemia	2	SVM	0.87	0.90	0.910	0.83	0.882
		Naive Bayesian	0.78	0.80	0.829	0.71	0.832
		KNN	0.82	0.88	0.873	0.83	0.746

Algorithm 1: Pseudocode for PCA

```
Produce an original solution Old_Config
Best_Fitness = Fitness (Old_Config)
For n=0 to # of iterations
   Perturbation ()
   If Fitness (New_Config) > Fitness (Old_Config)
      If Fitness (New_Config) > Best_Fitness
           Best_Fitness = Fitness (New_Config)
       End if
       Old_Config = New_Config
       Exploration ()
   Else
       Scattering ()
   End if
End for
Search ()
   For n=0 to # of iterations
```

```
    Small_Perturbation ()
     If Fitness (New_Config) > Fitness (Old_Config)
         If Fitness (New_Config) > Best_Fitness
             Best_Fitness = Fitness (New_Config)
         End if
         Old_Config = New_Config
      End if
Return
Scattering ()
   Pscattering = 1- (Fitness (New_Config) / Best_Fitness)
   If Pscattering > random (0,1)
       Old_Config = Random Result
   Else
       Search ()
   End if
Return
```

PCA takes into account high valued coordinated and neglects the data with low variance thus reducing the dimension of microarray datasets [28–30] (see Table 1a). Table 2 shows the details of feature extraction after PCA implementation.

Table 3 shows the bifurcation of data into training samples and test samples. The outcomes of are shown in Tables 4(a), 4(b) and 4(c) for Prostate Cancer and Colon Tumour and Leukemia data, respectively. Table 5 shows the confusion matrix for both datasets employing all classifiers. Support Vector Machine, Naive Bayes, and KNN classifiers are used to classify each dataset. The Precision, F1 score, AUC, Recall and Overall Accuracy are also calculated. For SVM-based classification, the polynomial kernel with C = 0.00010 and degree = 3 is used. Corresponding ROC for SVM is shown in Fig. 2 [39,40].

3.1.2. Feature selection using modified particle swarm optimization

The MPSO (Modified Particle Swarm Optimization) technique is employed at the second stage to pick features from high-dimensional microarray medical datasets like prostate cancer, leukemia, and colon carcinoma. Microarrays are being popularly referred as gene chip or DNA chip helps researchers by arming them with large number of genes at once, thus aiding the classification and prediction of a range of cancer data. However, these datasets exhibit a high level of dimensionality but are prone to be high-dimensional datasets are frequently unpredictable and off the mark. Particle swarm optimization (PSO), genetic algorithms (GA), and ant colony optimization (ACO) are just a few of the optimization techniques used by researchers to select effective feature subsets. After reduction the datasets are efficiently categorised with low cost of computation. In this work, MPSO is used to select features from leukemia, prostate cancer, and colon tumour using microarray datasets. Fig. 3 depicts the model used to classify microarray datasets using SVM, KNN, and Naive Bayes. Min-max technique is exercised to alter the original microarray data. The normalised microarray data is isolated from the training and testing data. Fig. 4 depicts the flowchart of MPSO [41–43].

3.2. MPSO (modified particle swarm optimization) for feature selection algorithm

In 1995, Kennedy and Eberhart created the PSO population-based evolutionary tool [44]. PSO was modelled on the social behaviour of flocking birds or schooling fish [45]. It is made up of a swarm of particles in most circumstances. The population consists of particles that wander through the issue space in search of the global optimal solution. Each particle has a vector of its current position and speed to direct its trajectory. Algorithm 2 shows the Modified Particle Swarm Optimization pseudocode.

Algorithm 2: Pseudocode of MPSO

```
Initialization
    For t=1 to X do
        Train each particle Y_i and authorize with CV
        Fitness calculation and complexity calculation of each particle
        Update pbest_i, pbest_{p, i} and the gbest
        if early stopping is fulfilled then
            return gbest
        end if
        Generation of new neighborhoods if gbest did not improve
        Update each lbest_i
        Update positions and velocities giving the formulas
        Transformation of features
        Limitations of velocities and out of range positions
End for
Return gbest
```

Equation (i) and (ii) from Table 6 above, respectively, show how a particle under contention i's velocity and location updates at iteration d. The areas where the best fitness has been discovered locally and globally are represented by the particles pbest and gbest, respectively.

As indicated in equation i, the new velocity vi is determined using the prior velocity, pbest and gBest. φ1 and φ2 are the social and local parameters, respectively. The range of values for the random variables rand1 and rand2 is [0, 1], and w is the weight factor of inertia.

Any particle represents a string Sn = [F1, F2, F3, Fn] in the modified PSO-based wrapper feature selection procedure, where n stands for the total number of features marked out. A portion of the chosen features serve to represent a particle's position. A smaller training set is produced by choosing traits from the original training set, and the classification performance of the training set is calculated. The chosen fitness function determines the particle's fitness.

MPSO approach is evaluated using KNN. To determine, the fitness function needs to be maximized for determining the particle's goodness.

The improved PSO method adjusts pbest, gBest, along with individual particle's velocity and location employing equations iii and iv from above Table 6, after all of the particles' fitness has been evaluated. The inertia weight, which is essential in maintaining equilibrium between exploration and exploitation [18], is used to compute the speed at

Table 5
Confusion Matrix for Prostate Cancer and Colon Tumour for different classifiers.

Dataset	Classifier	Confusion Matrix
Prostate Cancer	SVM	[[34 8] [0 25]]
	Naïve Bayesian	[[33 9] [0 25]]
	KNN	[[42 0] [18 7]]
Colon Tumor	SVM	[[10 3] [0 6]]
	Naïve Bayesian	[[10 1] [3 5]]
	KNN	[[10 1] [2 6]]
Leukemia	SVM	[[10 4] [0 8]]
	Naïve Bayesian	[[11 3] [3 5]]
	KNN	[[12 4] [0 6]]

which a particle's previous velocity contributes to its velocity at the current time step. Equations iii and iv are used as the main building blocks of the updated PSO to modify the ith particle's position and velocity in the d-dimensional space.

$\omega^d i$ is calculated using below equation vi in Table 6 above, where T_{max} is the number of the current iteration, and t is the total number of iterations. $\omega^d i$ decreases from 0.9 to 0.4 linearly and it is inversely proportional to t.

The MPSO-based wrapper feature mining method is used [31–33]. The suggested feature selection tool first asks the user how many key features they want to picked. Finally, the required number of features (best indices) is returned at the end of all iterations. As a result, the collected indices are used to create a new dataset. The smaller dataset is split into two sections: training and testing [34].

4. Experimental analysis

The study used three microarray medical datasets for leukemia, colon tumor, and prostate cancer. Results for a microarray dataset without feature selection and after using the MPSO feature selection approach are obtained, and they are displayed in Tables 7 and 8 respectively. The three classifiers are assessed using precision, recall, F-score, AUC, and overall accuracy on these datasets.

Following the selection of features from all datasets in the range of ten to one hundred, Table 8 provides information on the classification accuracy as well as the other important performance metric. Table 9 displays the top outcomes from all datasets with the most attributes chosen. In order to utilize features from datasets, the modified particle swarm optimization (MPSO) methodology is used in this study, highlighting the fact that classification accuracy varies depending on the quantity of features collected [35].

Table 9 displays the performance metrics for the SVM, KNN, and Naive Bayes classifiers for prostate cancer, colon cancer, and leukemia. These findings were achieved with the original normalised datasets without any characteristics from the datasets selected as shown in Fig. 5.

5. Conclusion and discussion

In this research, a novel Multi Class based Feature Extraction (MCFE) method has been proposed for medical data classification. Genomic datasets, or gene expression-based microarray medical datasets, are categorised for cancer diagnosis. The first stage involves applying a feature extraction technique. The Principal Component Analysis (PCA) is used to extract features for medical data classification to detect leukemia, colon tumors and prostate cancer. The MPSO (Modified Particle Swarm Optimization) technique is employed at the second stage to pick features from high-dimensional microarray medical datasets like prostate cancer, leukemia, and colon carcinoma. The classification of medical data is completed using SVM, KNN, and Naive Bayes classifiers.

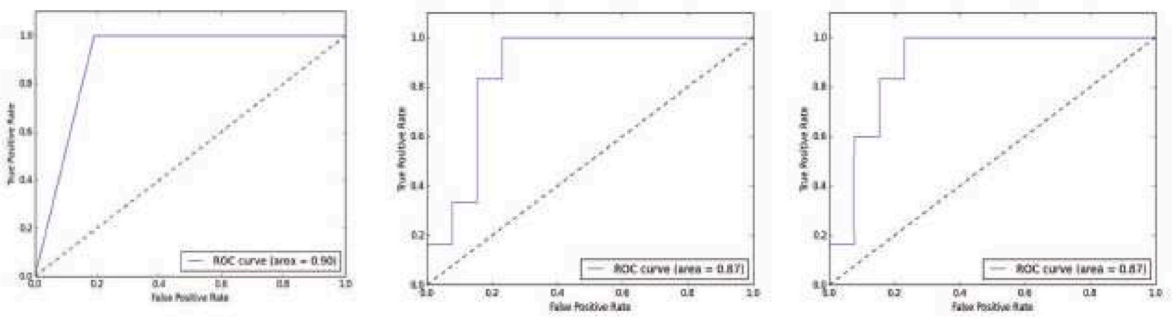


Fig. 2. ROC for the polynomial kernel with C = 0.00010 and degree = 3 is considered in SVM based classification. (a) Colon Tumor (b) Prostate Cancer (c) Leukemia.

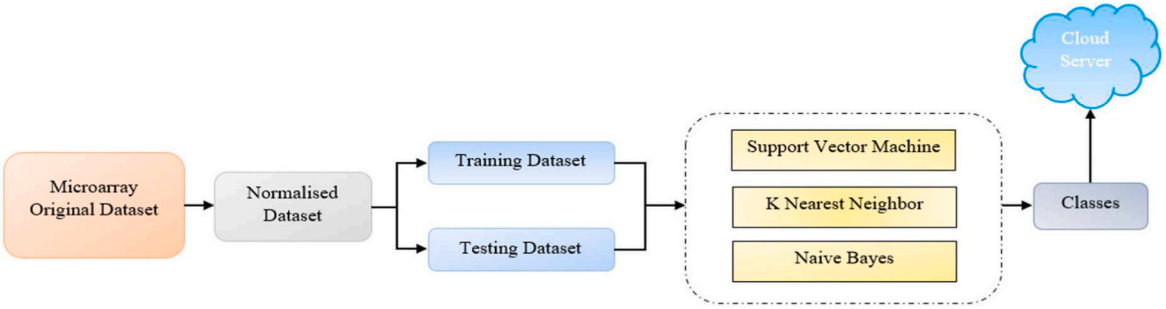


Fig. 3. Microarray data classification steps.

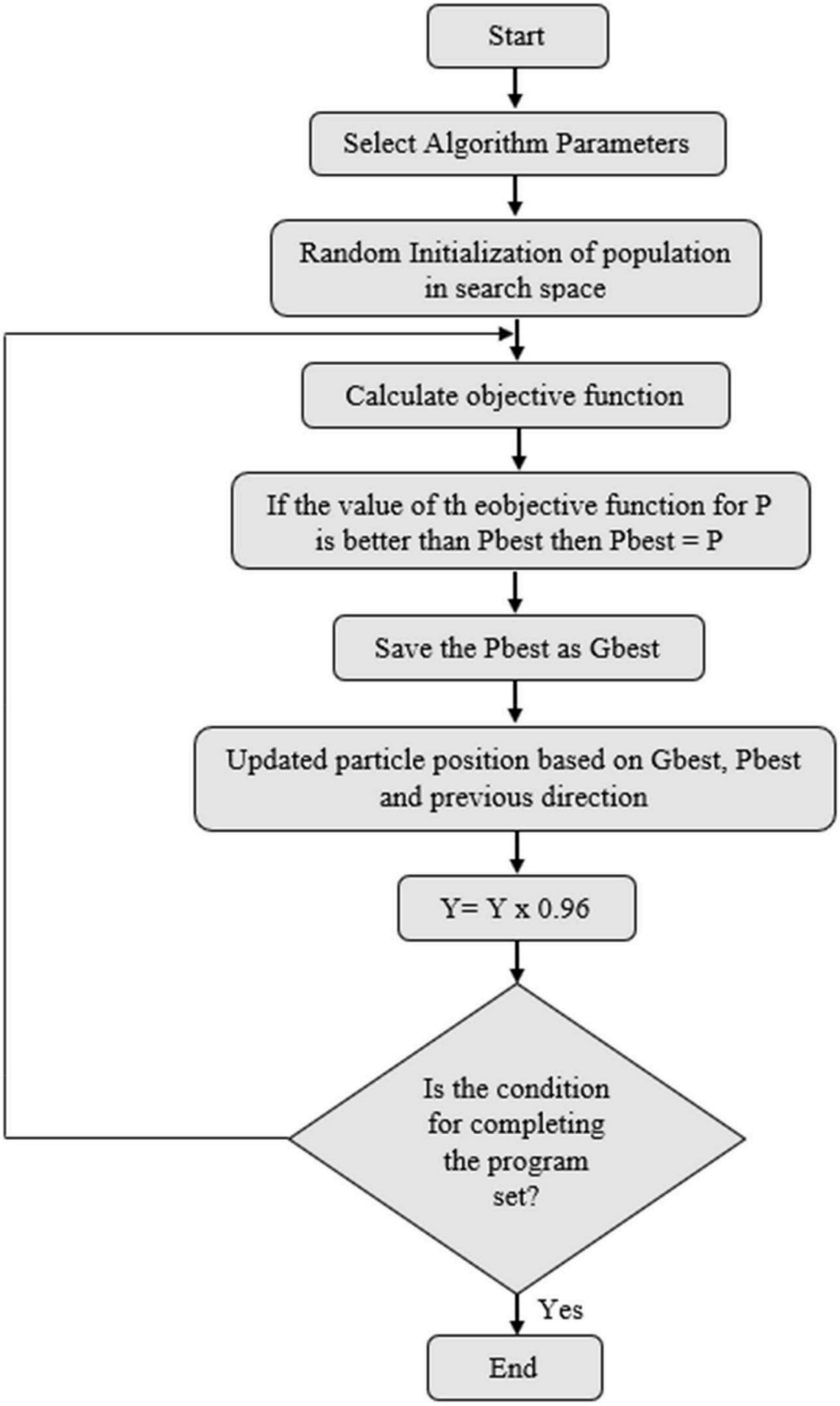


Fig. 4. Flowchart of MPSO

Table 6
Formulas used in MPSO.

Sr. No	Equations	Equation Nos
1	$v_i^{d+1}\omega.v_i^d + \varphi_1.rand_1(pBest_i - x_i^d) + \varphi_2.rand_2.(gBest_i - x_i^d)$	i
2	$x_i^{d+1} = x_i^d + v_i^{d+1}$	ii
3	$v_i^{d+1} = \eta[\omega_i^d.v_i^d + \varphi_1.rand_1.(pBest_i - x_i^d) + \varphi_2.rand_2.(gBest_i - x_i^d)]$	iii
4	$x_i^{d+1} = x_i^d + v_i^{d+1}$	iv
5	$\eta = \frac{2}{\lvert 2 - \varphi - \sqrt{\varphi^2 - 4\varphi}}$ WHERE φ=φ1 + φ2	v
6	$\omega_i^d = 0.9 - \frac{t}{T_{max}}$	vi

After that, the cloud IOT sensors are used to store the classified data. The suggested method outperforms the currently used approaches, providing better optimized outcomes for SVM, KNN, and Naive Bayes of 0.88, 0.86, and 0.73, respectively. The results of cancer data extraction and feature selection are also contrasted and assessed using certain performance measure factors. When comparing the classifiers based on AUC and overall accuracy scores, it is clear that SVM surpasses the others. In future work, it will develop various optimized algorithms for medical data classification and improve the accuracy.

Authorship contributions

Abdul Razzaque: Conceptualization and design of study, Acquisition of data, Analysis and/or interpretation of data, Drafting the manuscript, Revising the manuscript critically for important intellectual content. Dr. Abhishek Badholia: Conceptualization and design of study, Acquisition of data, Analysis and/or interpretation of data

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 9
Microarray Datasets based on optimally selected features (Best one).

Dataset	Classifier	Recall	Precision	AUC	F-Score	Classification Accuracy
Leukemia (nf = 10)	SVM	0.87	0.89	.90	0.86	96.00
	Naïve Bayesian	0.82	0.87	.87	0.84	92.00
	KNN	0.86	0.88	.89	0.89	98.00
Colon Tumor (nf = 10)	SVM	0.87	0.89	.90	0.86	100.0
	Naïve Bayesian	0.85	0.87	.90	0.89	97.00
	KNN	0.92	0.92	.90	0.91	100.00
Prostate Cancer (nf = 20)	SVM	0.92	0.94	.93	0.90	99.56
	Naïve Bayesian	0.81	0.88	.90	0.79	94.00
	KNN	0.89	0.92	.90	0.88	96.00

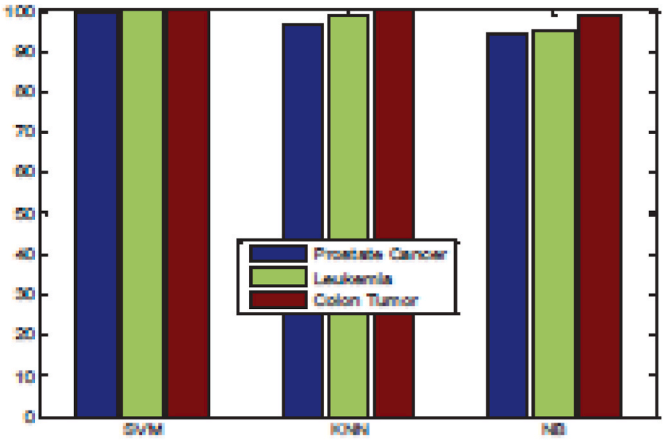


Fig. 5. Classification accuracy comparison for KNN, SVM, and Naive Bayes after applying feature selection using MPSO.

Table 7
Microarray datasets result before employing feature selection.

Dataset →	Prostate Cancer	Leukemia	Colon Tumor	Prostate Cancer	Leukemia	Colon Tumor	Prostate Cancer	Leukemia	Colon Tumor
Class. Algo. →	SVM	Naïve Bayesian	KNN	SVM	Naïve Bayesian	KNN	SVM	Naïve Bayesian	KNN
Recall	0.88	0.73	0.87	0.95	0.86	0.83	0.84	0.79	0.84
Precision	0.91	0.81	0.90	0.92	0.89	0.85	0.89	0.80	0.84
AUC	0.904	0.872	0.892	0.897	0.780	0.842	0.87	0.789	0.842
F-Score	0.88	0.68	0.87	0.92	0.86	0.84	0.85	0.78	0.84
Overall accuracy	0.880	0.865	0.731	0.924	0.882	0.832	0.884	0.801	0.750

Table 8
Classification accuracy got after applying MPSO feature selection.

Features	Leukemia			Colon Tumour			Prostate Cancer		
	NB	SVM	KKN	NB	SVM	KKN	NB	SVM	KKN
10	95.25	100.0	98.0	99.00	100.0	100.0	91.68	94.92	92.72
20	94.58	94.80	98.29	96.80	99.57	96.67	94.00	99.56	96.00
30	94.49	94.80	97.12	94.80	99.57	97.61	93.52	92.38	93.48
40	94.23	95.32	91.18	96.68	99.57	99.52	93.52	92.18	94.56
50	94.41	95.52	91.18	96.52	99.57	98.52	93.52	92.34	95.54
60	94.45	95.58	91.28	97.52	99.57	97.62	92.54	93.64	95.45
70	93.58	95.58	94.12	96.62	99.57	97.62	92.54	92.46	95.32
80	93.67	94.68	95.12	97.78	99.57	99.52	92.58	94.46	93.92
90	94.58	94.62	96.12	97.78	99.57	99.52	93.68	94.46	93.57
100	94.58	93.85	91.12	97.78	99.57	98.57	93.68	94.46	93.64

Data availability

Data will be made available on request.

References

[1] P. William, A. Shrivastava, N. Shunmuga Karpagam, T.A. Mohanaprakash, K. Tongkachok, K. Kumar, Crime analysis using computer vision approach with machine learning, in: N. Marriwala, C. Tripathi, S. Jain, D. Kumar (Eds.), Mobile Radio Communications and 5G Networks. Lecture Notes in Networks and Systems, vol. 588, Springer, Singapore, 2023, https://doi.org/10.1007/978-981-19-7982-8_25.

[2] G. Parmigiani, E.S. Garrett, R.A. Irizarry, S.L. Zeger, "The Analysis of Gene Expression Data: an Overview of Methods and Software", Springer, 2003.

[3] W.P. Kuo, E.-Y. Kim, J. Trimarchi, T.-K. Jenssen, S.A. Vinterbo, L. Ohno- Machado, A primer on gene expression and microarrays for machine learning researchers, J. Biomed. Inf. 37 (2004) 293–303.

[4] J. Dai Jian, L. Lieu, D. Rocke, et al., Dimension reduction for classification with gene expression microarray data, Stat. Appl. Genet. Mol. Biol. 5 (1) (2006) 1–21.

[5] P. William, N.K. Darwante, A.B. Pawar, M.A. Jawale, A. Verma, Framework for implementation of smart driver assistance system using augmented reality, in: N. Marriwala, C. Tripathi, S. Jain, D. Kumar (Eds.), Mobile Radio Communications and 5G Networks. Lecture Notes in Networks and Systems, vol. 588, Springer, Singapore, 2023, https://doi.org/10.1007/978-981-19-7982-8_30.

[6] S. Mishra, S. Choubey, A. Choubey, N. Yogeesh, J. Durga Prasad Rao, P. William, Data extraction approach using natural language processing for sentiment analysis, in: 2022 International Conference on Automation, Computing and Renewable Systems, ICACRS, Pudukkottai, India, 2022, pp. 970–972, https://doi.org/10.1109/ICACRS55517.2022.10029216.

[7] Paithankar Deepak Narayan, Pabale Abhijeet Rajendra, Vilas Kolhe Rushikesh, P. William, Madhukar Yawalkar Prashant, Framework for implementing air quality monitoring system using LPWA-based IoT technique, Measurement, Sensors (2023) 100709, https://doi.org/10.1016/j.measen.2023.100709. ISSN 2665-9174.

[8] K. Gupta, S. Choubey, Y. N, P. William, V.T. N, C.P. Kale, Implementation of motorist weariness detection system using a conventional object recognition technique, in: 2023 International Conference on Intelligent Data Communication Technologies and Internet of Things, IDCIoT, Bengaluru, India, 2023, pp. 640–646, https://doi.org/10.1109/IDCIoT56793.2023.10052783.

[9] Yawalkar Prashant Madhukar, Paithankar Deepak Narayan, Pabale Abhijeet Rajendra, P. William Rushikesh Vilas Kolhe, Integrated identity and auditing management using blockchain mechanism, Measurement: Sensors 100732 (2023), https://doi.org/10.1016/j.measen.2023.100732. ISSN 2665-9174.

[10] P. William, et al., Divination of air quality assessment using ensembling machine learning approach, in: 2023 International Conference on Artificial Intelligence and Knowledge Discovery in Concurrent Engineering, ICECONF, Chennai, India, 2023, pp. 1–10, https://doi.org/10.1109/ICECONF57129.2023.10083751.

[11] H.P. Varade, S.C. Bhangale, S.R. Thorat, P.B. Khatkale, S.K. Sharma, P. William, Framework of air pollution assessment in smart cities using IoT with machine learning approach, in: 2023 2nd International Conference on Applied Artificial Intelligence and Computing, ICAAIC, Salem, India, 2023, pp. 1436–1441, https://doi.org/10.1109/ICAAIC56838.2023.10140834.

[12] P. William Rushikesh Vilas Kolhe, Prashant madhukar yawalkar, deepak narayan paithankar, abhijeet rajendra pabale, smart city implementation based on internet of things integrated with optimization technology, Measurement: Sensors 27 (2023), 100789, https://doi.org/10.1016/j.measen.2023.100789. ISSN 2665-9174.

[13] M.A. Jawale, P. William, A.B. Pawar, Nikhil Marriwala, Implementation of number plate detection system for vehicle registration using IOT and recognition using CNN, Measurement, Sensors 27 (2023), 100761, https://doi.org/10.1016/j.measen.2023.100761. ISSN 2665-9174.

[14] P. William, G.R. Lanke, D. Bordoloi, A. Shrivastava, A.P. Srivastavaa, S. V. Deshmukh, Assessment of human activity recognition based on impact of feature extraction prediction accuracy, in: 2023 4th International Conference on Intelligent Engineering and Management, ICIEM, London, United Kingdom, 2023, pp. 1–6, https://doi.org/10.1109/ICIEM59379.2023.10166247.

[15] S.K. Korde, D.S. Rakshe, P. William, M.A. Jawale, A.B. Pawar, Experimental investigations on copper-based nanoparticles for energy storage applications, Journal of Nano-and Electronic Physics 15 (3) (2023).

[16] D.S. Rakshe, P. William, M.A. Jawale, A.B. Pawar, S.K. Korde, N. Deshpande, Synthesis and characterization of graphene based nanomaterials for energy applications, Journal of Nano-and Electronic Physics 15 (3) (2023).

[17] M.A. Jawale, A.B. Pawar, S.K. Korde, D.S. Rakshe, P. William, N. Deshpande, Energy management in electric vehicles using improved swarm optimized deep reinforcement learning algorithm, Journal of Nano-and Electronic Physics 15 (3) (2023).

[18] M. Rashmi, P. William, N. Yogeesh, D.K. Girija, Blockchain-based cloud storage using secure and decentralised solution, in: N. Chaki, N.D. Roy, P. Debnath, K. Saeed (Eds.), Proceedings of International Conference on Data Analytics and Insights, ICDAI 2023. ICDAI 2023. Lecture Notes in Networks and Systems, vol. 727, Springer, Singapore, 2023, https://doi.org/10.1007/978-981-99-3878-0_23.

[19] D.K. Girija, M. Rashmi, P. William, N. Yogeesh, Framework for integrating the synergies of blockchain with AI and IoT for secure distributed systems, in: N. Chaki, N.D. Roy, P. Debnath, K. Saeed (Eds.), Proceedings of International Conference on Data Analytics and Insights, ICDAI 2023. ICDAI 2023. Lecture Notes in Networks and Systems, vol. 727, Springer, Singapore, 2023, https://doi.org/10.1007/978-981-99-3878-0_22.

[20] P. William Neha Sharma, Kushagra Kulshreshtha, Gunjan Sharma, Bhadrappa Haralayya, Yogesh Chauhan, Anurag Shrivastava, Human resource management model with ICT architecture: solution of management & understanding of psychology of human resources and corporate social responsibility, JRTDD 6 (Aug. 2023) 219–230, 9s(2).

[21] P. William Nandita Tripathi, Gunjan Sharma, Bhadrappa Haralayya, K. Maheswari, Kushagra kulshreshtha, "is digital business growth in a green economy with environmental sustainability accelerate the organizational positive thinking and potential", JRTDD 6 (Aug. 2023) 209–218, 9s(2).

[22] K. Maheswari, P. William, Gunjan Sharma, Firas Tayseer Mohammad Ayasrah, Ahmad Y.A. Bani Ahmad, Gowtham Ramkumar, Anurag Shrivastava, Enterprise human resource management model by artificial intelligence to get befitted in psychology of consumers towards digital technology, JRTDD 6 (Sep. 2023) 209–220, 10s(2).

[23] G. Endait, A. Nikam, P. Bhorkade, D.G. Lokhande, B.S. Agarkar, P. William, Hybrid model to predict leaf disease prediction using ensembling machine learning approach, in: 2023 World Conference on Communication & Computing, WCONF, RAIPUR, India, 2023, pp. 1–7, https://doi.org/10.1109/WCONF58270.2023.10235119.

[24] P. William, S. Singh, Y. Kumar, A.K. Rao, A.K. Khan, D. Singh, Software reliability analysis with various metrics using ensembling machine learning approach, in: 2023 World Conference on Communication & Computing, WCONF, RAIPUR, India, 2023, pp. 1–6, https://doi.org/10.1109/WCONF58270.2023.10235066.

[25] P. William, D. Rajani, M. Gupta, R. Taluja, A.H.R. Alawadi, D.K. Yadav, Edge computing based traffic control management for distributed environment, in: 2023 World Conference on Communication & Computing, WCONF, RAIPUR, India, 2023, pp. 1–6, https://doi.org/10.1109/WCONF58270.2023.10235013.

[26] P. William, M. Gupta, N. Chinthamu, A. Shrivastava, I. Kumar, A.K. Rao, Novel approach for software reliability analysis controlled with multifunctional machine learning approach, in: 2023 4th International Conference on Electronics and Sustainable Communication Systems, ICESC, Coimbatore, India, 2023, pp. 1445–1450, https://doi.org/10.1109/ICESC57686.2023.10193348.

[27] P. William, M.A. Jawale, A.B. Pawar, S.K. Korde, D.S. Rakshe, N. Deshpande, Investigation of the electronic and optical properties of nanostructured glasses and composites, Journal of Nano-and Electronic Physics 15 (4) (2023).

[28] N. Deshpande, S.K. Sharma, P. William, A.B. Pawar, Empirical analysis of the emerging trends in the photovoltaic technologies of condensed matter physics, Journal of Nano-and Electronic Physics 15 (4) (2023).

[29] P. William, P.M. Yawalkar, D.N. Paithankar, A.R. Pabale, R.V. Kolhe, N. Deshpande, Green synthesis methods of nanostructures for environmental and biomedical applications, Journal of Nano-and Electronic Physics 15 (4) (2023).

[30] A.B. Pawar, S.K. Korde, D.S. Rakshe, P. William, M.A. Jawale, N. Deshpande, Analysis of silver nanoparticles as carriers of drug delivery system, Journal of Nano-and Electronic Physics 15 (4) (2023).

[31] P. William, G.S. Chhabra, A. Choubey, S. Choubey, S. Yuvaraj, Investigation of novel processes in the physics of condensed matter: recent advances and applications, Journal of Nano-and Electronic Physics 15 (4) (2023).

[32] A.R. Pabale, R.V. Kolhe, P. William, N. Deshpande, D.N. Paithankar, P. M. Yawalkar, Smart crack detection system using nanostructured materials with integrated optimization technology, Journal of Nano-and Electronic Physics 15 (4) (2023).

[33] A. Kumar, C. More, N.K. Shinde, N.V. Muralidhar, A. Shrivastava, C.V.K. Reddy, P. William, Distributed electromagnetic radiation based renewable energy assessment using novel ensembling approach, Journal of Nano-and Electronic Physics 15 (4) (2023).

[34] M. Chiranjivi, D. Obulesu, Ganesh Babu Loganathan, et al., J. Nano- Electron. Phys. 15 (4) (2023), 04020, https://doi.org/10.21272/jnep.15(4).04023.

[35] P. William, O.J. Oyebode, G. Ramu, M. Gupta, D. Bordoloi, A. Shrivastava, Artificial intelligence based models to support water quality prediction using machine learning approach, in: 2023 International Conference on Circuit Power and Computing Technologies, ICCPCT, Kollam, India, 2023, pp. 1496–1501, https://doi.org/10.1109/ICCPCT58313.2023.10245020.

[36] P. William, et al., Evolutionary survey on robotic process automation and artificial intelligence: industry 4.0, Robotic Process Automation (2023) 315–327, https://doi.org/10.1002/9781394166954.ch21.

[37] P. William, et al., Future of business organizations based on robotic process automation: a review, Robotic Process Automation (2023) 181–188, https://doi.org/10.1002/9781394166954.ch11.

[38] P. William, et al., Future of digital work force in robotic process automation, Robotic Process Automation (2023) 181–188, https://doi.org/10.1002/9781394166954.ch20.

[39] P. William, G. Ramu, L.R. Gupta, P. Sing, A. Shrivastava, A.P. Srivastava, Hybrid temperature and humidity monitoring system using IoT for smart garden, in: 2023 Second International Conference on Augmented Intelligence and Sustainable Systems, ICAISS, Trichy, India, 2023, pp. 1514–1518, https://doi.org/10.1109/ICAISS58487.2023.10250538.

[40] P. William, M.G. Yadav, S. Lakhanpal, A. Shrivastava, K.S. Rawat, S. Chaudhary, Artificial intelligence based performance models to support hydrologic rainfall conditions using ensembling approach, in: 2023 Second International Conference on Augmented Intelligence and Sustainable Systems, ICAISS, Trichy, India, 2023, pp. 223–229, https://doi.org/10.1109/ICAISS58487.2023.10250561.

[41] P. William, G.S. Chhabra, A. Kondekar, V.M. Tidake, S. Kulkarni, P.B. Narendra Kiran, Edge based web computing for traffic control management for distributed environment conditions, in: 2023 Second International Conference on Augmented

Intelligence and Sustainable Systems, ICAISS, Trichy, India, 2023, pp. 805–810, https://doi.org/10.1109/ICAISS58487.2023.10250563.
[42] V. Pawar, S. Kumawat, M. Ahire, R. Musmade, R.R. Nikam, P. William, ETL based billing system for azure services with cost estimation using cloud computing, in: 2023 Second International Conference on Augmented Intelligence and Sustainable Systems, ICAISS, Trichy, India, 2023, pp. 1163–1166, https://doi.org/10.1109/ICAISS58487.2023.10250628.
[43] A. Shimpi, R. Gambhir, A. Diwate, S. Udawant, N.Y. Siddiqui, P. William, Framework for assessment of bitcoin price prediction using ensembling machine learning approach, in: 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN, Salem, India, 2023, pp. 349–353, https://doi.org/10.1109/ICPCSN58827.2023.00063.
[44] P. William, G. Ramu, L. Kansal, P.P. Patil, A. Alkhayyat, A.K. Rao, Artificial intelligence based air quality monitoring system with modernized environmental safety of sustainable development, in: 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN, Salem, India, 2023, pp. 756–761, https://doi.org/10.1109/ICPCSN58827.2023.00130.
[45] S.C. Bhangale, H.P. Varade, S.R. Thorat, D.K. Roy, P. William, A. Verma, Air quality hotspot monitoring with trajectories of IoT in smart city implementation, in: 2023 3rd International Conference on Pervasive Computing and Social Networking, ICPCSN, Salem, India, 2023, pp. 1274–1279, https://doi.org/10.1109/ICPCSN58827.2023.00214.